Kian Javaheri

kianjavaheri911@gmail.com | Portfolio | GitHub | LinkedIn

Education

Arizona State University Aug. 2022 – May 2026
B.S. in Computer Science, Economics *Tempe, AZ*

- **GPA: 3.93** · Summa Cum Laude | Barrett, The Honors College

Experience

Sandia National Laboratories May 2023 – Aug. 2024
R&D Bioresource and Environment Security Intern *Livermore, CA*

- Led and designed a standalone **optimization tool in Python** that ingests operational data across multiple bioresource systems and maximizes projected cash flow via configurable optimization algorithms
- Created and validated shapefile datasets, then engineered a companion **QGIS plugin** in Python that connects to the optimization tool via an internal API, letting users gauge how different feedstock locations affect profit yields

Hanemann Lab @ Arizona State University May 2025 – May 2026
Junior Researcher *Tempe, AZ*

- Automated large-scale document retrieval from the Arizona Department of Water Resources using **Selenium WebDriver**, eliminating manual collection across hundreds of annual reports
- Engineered **OCR extraction** pipelines to parse and structure tabular data from scanned PDFs, producing clean, analysis-ready datasets for downstream econometric modeling

Selected Projects

Revolution Parts Shipping Service with Caching | *React, PostgreSQL, Flask, Docker*

- Architected a high-throughput relational caching layer using **PostgreSQL** and **Flask** within a **Dockerized** microservice environment, intercepting and caching external carrier API responses to reduce redundant network calls
- Achieved a **64.4% reduction in processing latency** (modal response time: 334ms → 119ms), delivering a **2.8x speedup** over live API calls and substantially reducing data-layer overhead

Santa Cruz Rental Price Model | *Python, LightGBM, Linear Regression*

- Trained a **LightGBM** rent model for studios and one-bedrooms on 1,765 listings, cutting error from a 15.2% median baseline to a **12.6% MAPE** on 783 held-out listings

Material Boxes | *Java, Fabric, Claude API*

- Developed a client-side **Fabric mod** for Minecraft that tracks materials needed for a build, supporting import via pasted text, a Litematica export, or a screenshot parsed with **OCR** through the Claude API
- Released the mod publicly on **CurseForge**, reaching **70+** user downloads

Leadership & Activities

Arizona State University
Teaching Assistant, CSE310: Data Structures & Algorithms *Tempe, AZ*

- Led exam review sessions on **red-black trees**, **dynamic programming**, and **minimum spanning tree** algorithms, and held 5+ office hours per week guiding students through algorithm design challenges

Google Data Analytics Certificate
Coursera

- Completed all 8 courses including data cleaning and **SQL querying** in **R**

Technical Skills

Machine Learning & AI: Gradient Boosting, Ridge, Linear Regression, Fine-Tuning, Predictive Modeling
Languages: Java, Python, C/C++, SQL, JavaScript/TypeScript, HTML/CSS, R
Frameworks: React, Node.js, Flask, JUnit
Developer Tools: Git, GitHub, Docker, JupyterHub, Firebase, MATLAB, QGIS, Postman, Stata, Qualtrics
Libraries: PyTorch, LightGBM, Scikit-learn, Pandas, NumPy, Matplotlib, Selenium